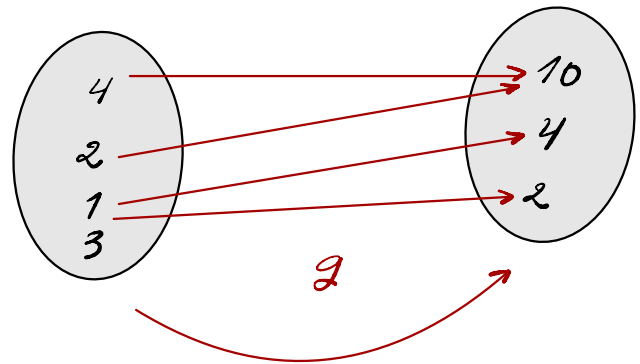
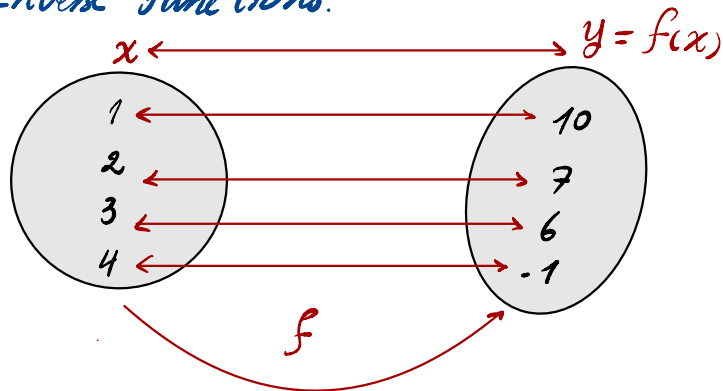


Lecture 1. Jan 20 - 2026.

6.1. Inverse Functions and Their Derivatives.

* Inverse Functions.

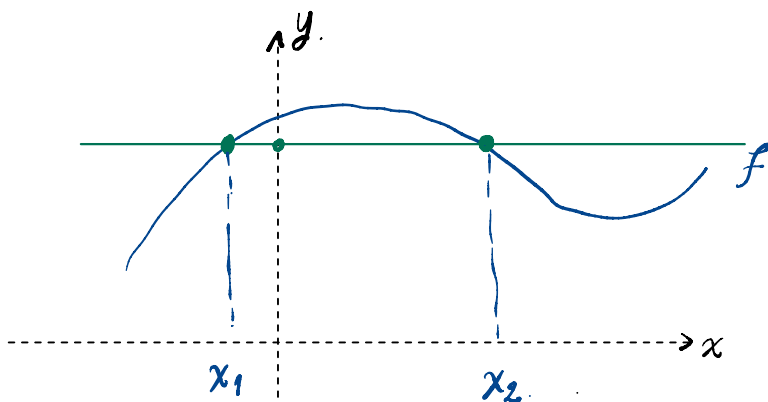


Not all functions possess inverses.

Def. A function f is called a one-to-one function if it never takes on the same value twice; that is,

$$f(x_1) \neq f(x_2) \text{ whenever } x_1 \neq x_2.$$

Horizontal line test.

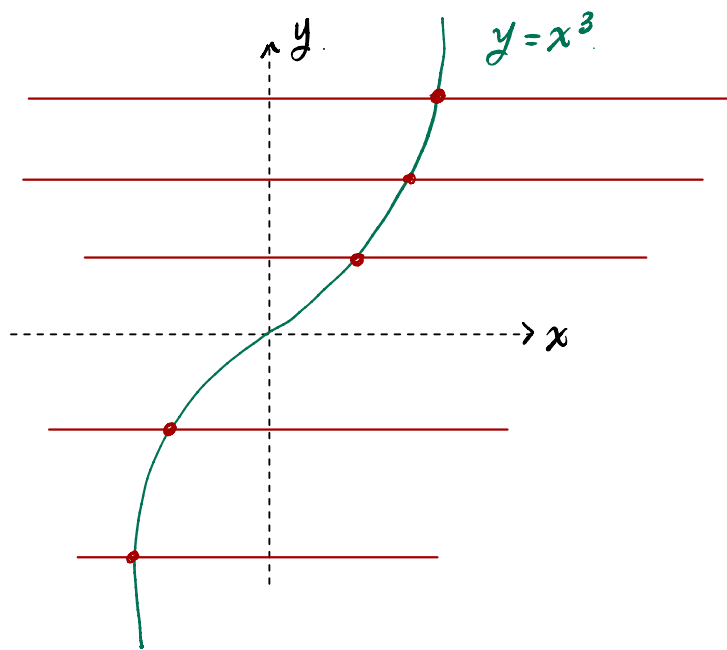


A function is one-to-one if and only if no horizontal line intersects its graph more than once.

Example. Is the function $f(x) = x^3$ one-to-one?

Sln. If $x_1 \neq x_2$, then $x_1^3 \neq x_2^3$ (two different numbers can't have the same cube). Therefore, by Definition $f(x) = x^3$ is one-to-one.

Sln. We know.



We see that no horizontal line intersects the graph of $f(x) = x^3$ more than once. Therefore, by Horizontal line test, f is one-to-one.

Example: is the function $g(x) = x^2$ one-to-one?

Sln. This function is not one-to-one b/c.

$x_1 = 2$, $x_2 = -2$ are different but

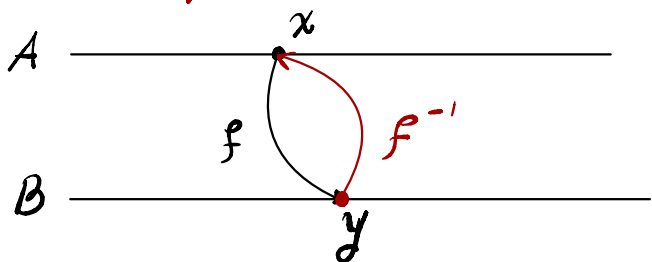
$$f(x_1) = (2)^2 = 4 \quad \text{and} \quad f(x_2) = (-2)^2 = 4$$

$$\text{i.e., } f(x_1) = f(x_2)$$

Def. Let f be a one-to-one function with domain A and range B . then its inverse function f^{-1} has domain B and range A and is defined by

$$f^{-1}(y) = x \quad \text{if } f(x) = y.$$

Rmk. This definition says that if f maps x into y , then f^{-1} maps y back into x .



$$\text{Dom } f^{-1} = \text{Range } f$$

$$\text{Range } f^{-1} = \text{Dom } f$$

For example, the inverse function of $f(x) = x^3$ is $f^{-1}(x) = x^{1/3}$ because if $y = x^3$ then

$$f^{-1}(y) = f^{-1}(x^3) = (x^3)^{1/3} = x$$

⚠ Do not mistake the -1 in f^{-1} for an exponent. Thus,

$f^{-1}(x)$ does not mean $\frac{1}{f(x)}$.

Example. If f is one-to-one function and $f(1) = 5$, $f(3) = 7$, $f(8) = -10$ find $f^{-1}(7)$, $f^{-1}(5)$, $f^{-1}(-10)$.

Sln.

$$f^{-1}(7) = 3 \quad \text{b/c} \quad f(3) = 7$$

$$f^{-1}(5) = 1 \quad \text{b/c} \quad f(1) = 5$$

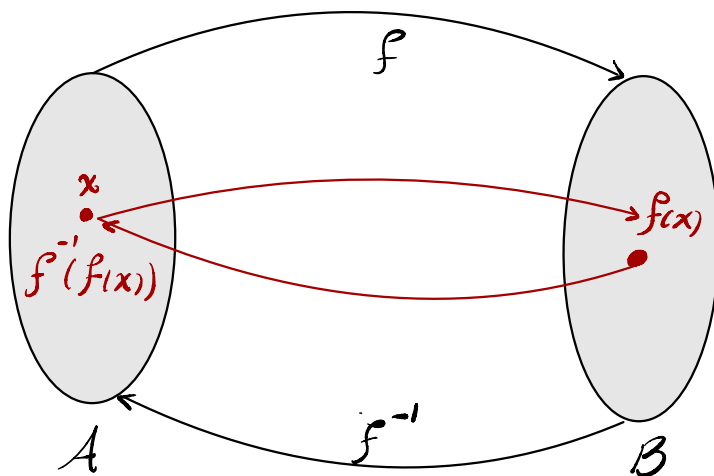
$$f^{-1}(-10) = 8 \quad \text{b/c} \quad f(8) = -10.$$

$$f^{-1}(x) = y \Leftrightarrow f(y) = x$$

We have cancellation equations.

$$f^{-1}(f(x)) = x \quad \text{for every } x \in \text{Dom } f = A.$$

$$f(f^{-1}(x)) = x \quad \text{for every } x \in \text{Range } f = B.$$



for example if $f(x) = x^3$, $f^{-1}(x) = x^{1/3}$

$$f^{-1}(f(x)) = f^{-1}(x^3) = (x^3)^{1/3} = x.$$

$$f(f^{-1}(x)) = f(x^{1/3}) = (x^{1/3})^3 = x.$$

How to find the inverse function of a one-to-one function?

Step 1. write $y = f(x)$

Step 2. Solve the equation for x in terms of y (if possible).

Step 3. To express f^{-1} as a function of x , interchange x and y the resulting equation is $y = f^{-1}(x)$.

Example. Find the inverse function of $f(x) = x^3 + 2$.

Sln. $f(x)$ is one-to-one.

Step 1. $y = f(x) \Leftrightarrow y = x^3 + 2$

Step 2. $y = x^3 + 2$

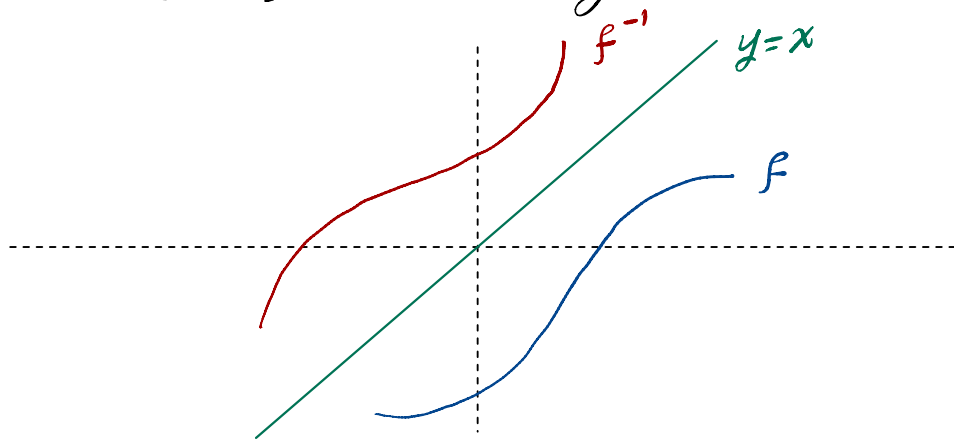
$$y - 2 = x^3$$

$$\sqrt[3]{y-2} = x.$$

Step 3. $y = \sqrt[3]{x-2}$

then $f^{-1}(x) = \sqrt[3]{x-2}$.

Note that $\text{Dom } f^{-1} = \mathbb{R} = \text{Range } f$.



Remk. The graph of f^{-1} is obtained by reflecting the graph of f about the line $y=x$.

* Derivatives of inverse functions.

Thm. If f is one-to-one continuous function defined on an interval, then the inverse f^{-1} is also continuous.

Thm. If f is one-to-one differentiable function with inverse f^{-1} and $f'(f^{-1}(a)) \neq 0$, then the inverse function is differentiable at a and

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}.$$

Pf.

We know that $f(f^{-1}(x)) = x$ for every x .

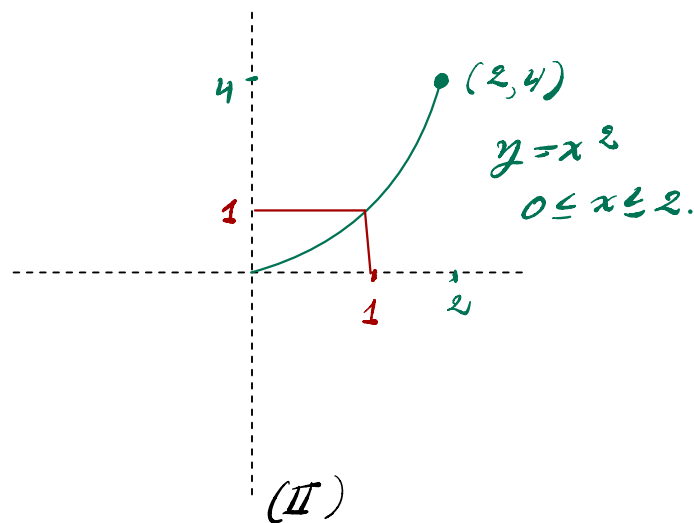
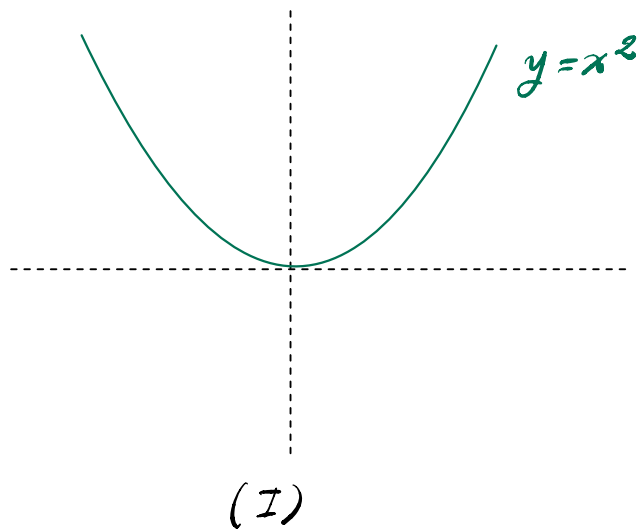
By implicit derivatives.

$$\frac{d}{dx} [f(f^{-1}(x))] = \frac{d}{dx} (x)$$

$$f'(f^{-1}(x)) \cdot \frac{d}{dx} (f^{-1}(x)) = 1$$

$$\frac{d}{dx} (f^{-1}(x)) = \frac{1}{f'(f^{-1}(x))}$$

Example. Although the function $y=x^2$, $x \in \mathbb{R}$ is not one-to-one and therefore does not have an inverse, we can turn it into one-to-one function.



in the case (II) we have a one-to-one function. without computing a formula for $(f^{-1})'$ we can still calculate $(f^{-1})'(1)$
 $f(1)=1$ then $f^{-1}(1)=1$ since $f'(x)=2x$ by the thm

$$(f^{-1})'(1) = \frac{1}{f'(\underline{f^{-1}(1)})} = \frac{1}{f'(\underline{1})} = \frac{1}{2}.$$

for you: $(f^{-1})'(1)$ where $f(x) = 2x + \cos x$.